(Please write your Enrollment Number) Enrollment No:_____

MID-TERMEXAMINATION
(CBCS)
B.Tech (CSE/IT/MAE/ECE/DMAM/CSEAI/ECEAI), Semester: 3
(Oct, 2022)

Subject Code: BCS 201 Subject: Data Structures
Time : 1 ½ Hours Maximum Marks : 30
Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 (2.5*4)

(a) Write an algorithm/program to count the number of zeros in a two dimensional array. Analyse the time complexity of the algorithm
(b) What is the drawback of linear array implementation of queue? Suggest a solution.
(c) Convert the following infix expression to postfix expression using stack {(A+B)*((C-D)/(E*F))}
(d) Write an algorithm/program to print a linked list in reverse order?

Q2 (5,5)

(a) Write an algorithm/program to swap the second biggest and second smallest element in an array
(b)Write an algorithm/program to swap two adjacent nodes (node at pointer P and its next node) in a doubly linked list. Analyze the time complexity of the algorithm.

Q3 (5,5)

(a) Write an algorithm/program to split a linked list into two by manipulating the links. The first list contains all odd nodes and the second list contains all even nodes.
(b) Write an algorithm/program to return n^{th} node from the end of a linked list

Q4 (5,5)

(a) Implement (push and pop operations)two stacks in a common array where both stacks grow in opposite direction.
(b) Implement (insertion and deletion operations) a queue using singly linked list